higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N5

(8080365)

26 November 2019 (X-Paper)
09:00–12:00

Calculators may NOT be used.

This question paper consists of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Keep subsections of questions together.
5. Give calculation processes and calculated answers in THREE fractional radix spaces, for example $10{,}101_2$.
6. Do ALL sketches neatly with a pencil and a ruler – NO freehand lines.
7. Use only a black or blue pen.
8. Write neatly and legibly.

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QUESTION 1

Convert each of the following numbers to its binary equivalent and complete the calculation in the binary number system (follow the instructions in brackets):

1.1 $23,5_{10} - C,4_{16}$

(Use the two's-complement method and convert the answer to hexadecimal.) (6)

1.2 $34,25_{10} \times 107,5_{8}$

(Convert the answer to decimal.) (6)

1.3 $132,5_{8} \div 21,A_{16}$

(Convert the answer to octal.) (6)
[18]

QUESTION 2

Design and draw a majority-voter circuit consisting of THREE variables A, B and C. The output (y) of this circuit will always agree with the majority of the inputs. Use a truth table and three-variable Karnaugh map. **[20]**

QUESTION 3

3.1 Sketch a two-input complementary C-MOS NAND gate. Show the input and output connections. (6)

3.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (3.2.1–3.2.5) in the ANSWER BOOK.

3.2.1 Fan out is the maximum number of inputs that can be served by an output.

3.2.2 Wiring together the outputs of an open collector circuit is called adding action.

3.2.3 Tristate logic is used to interface digital devices such as memories to a bus.

3.2.4 The TTL family uses field-effect transistors.

3.2.5 The outputs of a TTL circuit should never be connected together.

(5 × 2) (10)
[16]

QUESTION 4

4.1 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–F) next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
4.1.1	Time required to position the read/write head over the cylinder with data to be accessed	A	latency time
4.1.2	When data is being accessed and rotational delay is as much as one complete revolution	B	disc memories
4.1.3	Contraction of modulator/demodulator	C	seek time
4.1.4	Information stored in a concentric format and coated with a magnetisable surface	D	drum memories
4.1.5	Information stored in parallel tracks on movable metallic cylinders coated with a magnetised surface	E	modem
		F	UART

(5 × 1) (5)

4.2 Give FOUR examples of nonimpact printers. (4)

4.3 What is the function of UART? (2)
[11]

QUESTION 5

5.1 Draw a block diagram of a computer organisation and show ALL links amongst the unit. (10)

5.2 Name FIVE peripheral output devices commonly used. (5)
[15]

QUESTION 6

6.1 Sketch a block diagram of USART. (9)

6.2 Give ONE disadvantage of RAM. (1)
[10]

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QUESTION 7

7.1 Draw an equality comparator (XNOR-gate) by using NOR-gates. (5)

7.2 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (7.2.1–7.2.5) in the ANSWER BOOK.

7.2.1 $\mathrm{AB} + \mathrm{A}$:

A AB
B 1
C 0
D A

7.2.2 $\mathrm{R} + \mathrm{S} + \overline{\mathrm{RS}}$:

A R
B 1
C $\mathrm{R} + \mathrm{S}$
D RS

7.2.3 $\overline{\mathrm{A}\overline{\mathrm{B}} + \overline{\mathrm{A}}\mathrm{B}}$:

A 1
B A
C 0
D $\overline{\mathrm{A}}\overline{\mathrm{B}} + \mathrm{AB}$

7.2.4 $(x + y)(x + z)$:

A 1
B x
C $x + yz$
D $x^2 + xy + yz + z^2$

7.2.5 $\mathrm{AB}(\mathrm{C} + \overline{A}\,\overline{B})$:

A 1
B A
C 0
D ABC

(5 × 1) (5)
[10]

TOTAL: 100

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